

Front Page

Nomu FlowCart Business Plan

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Business Plan

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01 Executive Summary

Nomu FlowCart is developing a modular robotic housekeeping logistics system designed to improve operational efficiency in hotels. Housekeeping remains one of the largest cost centers in the hospitality industry, with labor costs averaging about \$44 per occupied room in the U.S., representing roughly \$58 billion annually in the U.S. and about \$190 billion globally. Despite increasing adoption of AI in hospitality, housekeeping workflows remain largely manual and operationally inefficient.

FlowCart transforms the traditional housekeeping cart into an intelligent robot cart system. The platform combines a smart cart hub with detachable robotic modules that independently transport clean supplies, dirty linens, and waste across hotel floors. By distributing material movement across smaller mobile units, FlowCart reduces unnecessary trips, lowers physical strain on housekeeping staff, and improves room turnover efficiency. The system also provides managers with real-time visibility into cleaning progress and supply usage.

Nomu operates under a Robot-as-a-Service (RaaS) model with a \$2,000 setup fee and \$350 monthly subscription per robot, allowing hotels to adopt the system with minimal upfront investment while generating recurring revenue for the company. Financial projections estimate 700 robots deployed by Year 5, generating approximately \$5.6 million in annual revenue, with break-even reached at around 450 deployed robots.

Nomu is currently seeking early funding to support prototype development and pilot deployments. Initial capital will come from a \$25,000 business plan competition prize, followed by planned pre-seed, seed, and Series A funding rounds to support product development and scaling across hotel networks.

02 Our Product: Nomu FlowCart



FlowCart is a modular robotic housekeeping logistics system designed to support the movement of materials throughout hotel floors. At the center of the system is a smart cart platform that coordinates detachable robotic modules for linens, supplies, waste, and cleaning tools.

Unlike traditional housekeeping carts, which uniformly transport supplies regardless of material category or associated workflow, FlowCart distributes tasks across multiple mobile modules. These modules can independently transport clean supplies to housekeepers

and return dirty linens or trash to service rooms. By separating material flows into smaller units, the system reduces hallway congestion and eliminates many back-and-forth trips that slow room turnover. By reducing these repetitive tasks, housekeepers can focus on more complex tasks that enhance the guest experience.

Beyond material movement, FlowCart also functions as an operational coordination platform. The system monitors supply usage, tracks cleaning progress, and supports smarter scheduling of housekeeping tasks across floors. By combining modular robotics with workflow intelligence, FlowCart transforms the housekeeping cart from passive equipment into an intelligent logistics infrastructure for hotel operations. We have now finished building our initial prototype (see the image above).

03 Market Analysis

To seek market opportunities, we conducted both customer discovery research and second-hand data collection.

Customer Discovery: Operational Pressure in Housekeeping

Our primary customer discovery involved on-site interviews and operational observations at the Statler Hotel as a deeper investigation. Statler is a mid-sized property with about 150 rooms and a unionized workforce, representing operational conditions common in upper-midscale and upscale hotels. Interviewees included 1) Milo, the Housekeeping & Guest Client Supervisor, 2) Aaron, the Manager of Hotel Training, and 3) Shawn, the Purchasing Agent.

Across departments, interviews revealed consistent operational challenges:

1. Cost pressure and efficiency

Labor represents approximately 35-40% of Statler's operating costs, making it the hotel's largest expense. Recent union contracts increased wages by 8% without proportional increases in room rates or food prices. At the same time, the hotel must meet an annual profit target of roughly \$3 million, with the proceeds inclusive of this threshold returned to the Johnson School of Business. These pressures make operational efficiency, particularly in housekeeping, the most labor-intensive department, critical to maintaining profitability.

2. Physical strain and staffing pressure

The most urgent challenge identified was the physical toll of housekeeping work. Milo explained that housekeepers are expected to complete approximately 14 rooms per day, about 30 minutes per room. When staffing is short—which he described as common—supervisors often step in to clean rooms themselves. As Milo noted:

“They're human. They can't do it physically if they don't have the time.”

Shoulder injuries are the most common ergonomic issue in the department, despite ongoing ergonomic training and stretching programs; this highlights the physical demands of the job and the need for tools that reduce repetitive strain. FlowCart directly addresses this problem by reducing cart pushing, supply-carrying trips, and repetitive bending through electric assist and modular supply organization.

3. Existing tools and workflow limitations

While current apps help manage operational tasks at a high level, they provide limited visibility into real-time housekeeping workflows. Supervisors still rely on manual inspections and direct observation to track progress and maintain quality. On the logistics side, purchasing operations use systems such as Notifii for package tracking and Attico for food ordering, supplemented by email and text communication with vendors. Across departments, the pattern is clear: tools are adopted when they integrate naturally into existing tasks and workflows, and resisted when they add additional complexity. FlowCart is designed to align with this reality by embedding technology into a tool staff already use: the housekeeping cart.

Why Customers Are Seeking Change

1. Labor costs continue to rise faster than hotel revenues can absorb, increasing pressure on operational efficiency.

2. Labor supply is becoming less predictable. Increased immigration enforcement and stricter employment verification policies are contributing to staffing shortages in the hospitality industry, particularly in housekeeping roles that have historically relied on immigrant labor. As hiring becomes more difficult, hotels must find ways to maintain productivity with fewer staff.

3. Staffing shortages during high-occupancy periods are a recurring operational condition, not an exception. Supervisors frequently manage these situations through manual intervention, guest room substitutions, and direct participation in cleaning tasks. Technologies that improve each housekeeper's productivity per shift can deliver immediate operational value.

4. Physical demands of housekeeping create both injury and retention risks. Long-tenured employees carry valuable institutional knowledge, yet shoulder and back injuries are common in the profession. Ergonomic improvements that reduce physical strain can help protect staff health while maintaining operational continuity.

Market Opportunities

The housekeeping market within the hospitality market is growing. The United States has approximately 5.3 million hotel rooms, generating over 1.3 billion occupied room nights annually¹, while the global market includes 17-18 million rooms. The

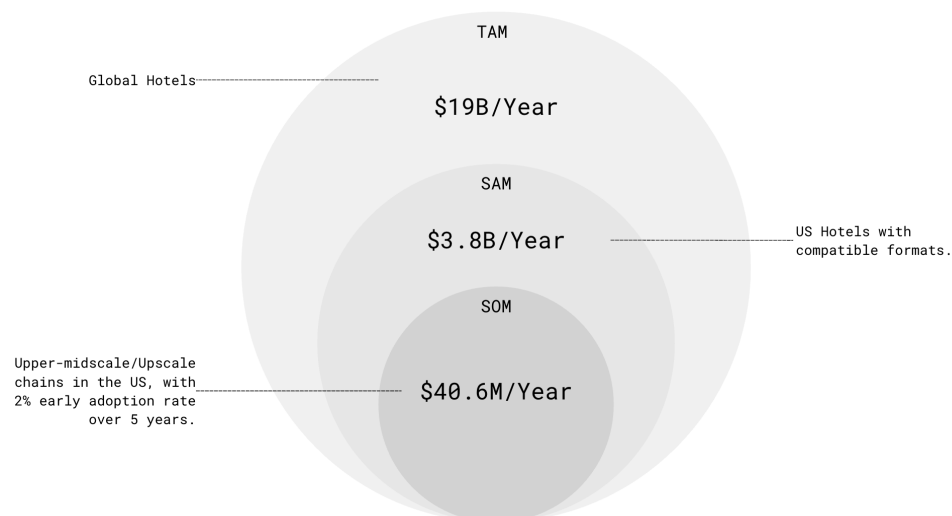
¹ American Hotel & Lodging Association (AHLA), *2025 Economic Impact of the U.S. Hotel Industry*

housekeeping cart market is valued at approximately \$650 million and is projected to grow at a 10.1% CAGR from 2026 to 2033².

Housekeeping is a major cost center: Housekeeping labor costs average \$44-\$45³ per occupied room in the U.S., representing approximately \$58 billion in domestic labor spending annually and about \$190 billion globally. Improving efficiency in this area creates immediate financial value for hotel operators.

Hospitality is embracing new technologies, signalling the right time to enter the market. 98% of hotel owners report using AI in some aspect of operations. 78% of hotel chains have adopted AI technologies. 76% of hotel executives believe AI will fundamentally change the industry. Operational departments remain underserved: most hotel AI adoption has focused on guest-facing systems, leaving large operational functions like housekeeping relatively untapped for automation.

Predicted FlowCart Market Size



The FlowCart market is expected to grow alongside the hospitality industry, with an estimated 6-8% CAGR for the total addressable market (TAM), 8-10% CAGR for the serviceable available market (SAM) driven by hospitality technology adoption, and 15-25% CAGR for the serviceable obtainable market (SOM) reflecting early-stage growth and expansion of automation solutions in hotel operations⁴.

² According to Prognos One report for Housekeeping Cart Market Forecast 2026 - 2033.

³ According to the HotelData by actabl report on hotel budget.

⁴ According to Grand View Research US Hotel Market Report.

04 Go-to Market Strategy

Target Customers Market

FlowCart's primary customers are hotel owners, general managers, and operations directors who are responsible for housekeeping performance, labor costs, and service quality. Our initial focus is on mid- to large-scale hotels and hotel chains where room turnover speed, staffing efficiency, and operational visibility have a direct financial impact.

Nomu FlowCart's first customers are innovation-oriented hotel operators with both the operational complexity and the organizational flexibility to quickly pilot new technologies. We identified three categories of potential early adopters:

Independent boutique hotels and owner-operated lifestyle properties. These operators typically have centralized decision-making at the ownership or property level, allowing them to pilot operational technologies without navigating the lengthy approval processes common within large hotel chains. Industry reporting from PhocusWire has noted that independent hotel operators frequently adopt new technologies faster than branded chains, particularly when the tools offer clear operational efficiency gains.

Soft-brand and independent collections affiliated with major hotel companies, such as Marriott's Autograph Collection, Hilton's Curio Collection, and Hyatt's Independent Collection brands. These properties benefit from the marketing and distribution systems of major hotel groups while retaining greater operational flexibility at the property level. As a result, they are often able to test new operational technologies at individual properties before broader brand-wide adoption.

Tech-forward lifestyle hotel brands also represent a strong early adoption segment. Brands such as **Moxy, Andaz, and YOTEL** have built their identities around modern, technology-enabled guest experiences and operational experimentation. Several hotels within these segments have already piloted robotics solutions for operational tasks.

Sales Channels

We will sell B2B at **industry conferences, trade exhibitions, hospitality technology media, and professional networks.** These channels act as key marketplaces where hotel owners, general managers, and operations leaders evaluate new vendors and pilot emerging technologies. For example, major hospitality technology conferences include HITEC (Hospitality Industry Technology Exposition & Conference) or The Lodging Conference, The Hospitality Show, and BDNV.

Hotel operators also rely on hospitality technology, media platforms, and peer networks when researching new solutions. Publications such as *Hotel Tech Report*, *PhocusWire*, and *Hospitality Net* influence purchasing decisions through product reviews and case studies. At the same time, adoption often spreads through peer recommendations within

hotel ownership groups and management companies, where successful pilot deployments at one property can lead to expansion across multiple hotels.

Revenue Model

We sell directly to hotels and hotel groups through a B2B sales model, starting with pilot deployments in selected properties to demonstrate ROI. Successful pilots are expanded into multi-property or chain-wide rollouts. Partnerships with hotel groups, facility management providers, and hospitality technology integrators will support scaling.

1. **Hardware sales or leasing** of FlowCart units and core modules (RaaS)
2. **Recurring software subscription** for management features such as task planning, real-time tracking, analytics, and system updates.
3. **Add-on modules and accessories** (e.g., specialized modules for deep cleaning, premium properties, or high-turnover operations).

This hybrid model provides upfront revenue from deployments and recurring, high-margin revenue from software and services.

Pricing is benchmarked against the cost of labor hours saved and improved room turnover speed. For most hotels, the system is designed to pay for itself through efficiency gains and labor cost savings within a defined payback period.

We will charge two types of fees:

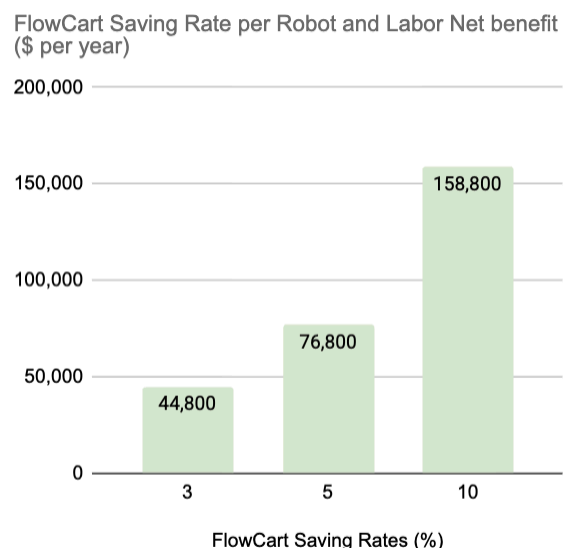
1. New robot setup fee: \$2,000. It covers system installation, configuration of the FlowCart modules, onboarding of hotel staff, and integration with the hotel's operational workflows.
2. RaaS subscription, \$350/month.

Benefit to Hotels

FlowCart's value proposition is grounded in measurable operational improvement. The following KPI framework connects FlowCart's core capabilities to specific, trackable outcomes in hotel housekeeping operations.

1. Room Turnover Time (Minutes per Room)

Target improvement: Reduce average per-room turnover time by 10-15% (approximately 3-4 minutes per room).



The operational baseline at Statler is approximately 30 minutes per room per checkout clean, with a daily quota of 14 room credits per housekeeper. Industry data from our interviews and benchmarks confirms this is a representative standard across upper-midscale and upscale properties.

2. Labor Cost per Occupied Room

The industry benchmark for housekeeping labor cost is approximately \$44.62 per occupied room in the U.S. (per STR/AHLA data, as cited in the Market Analysis section). Even a small improvement in operational efficiency can generate significant cost savings for hotels. For a typical 150-room property at 67% occupancy, a 10% savings rate can generate a net benefit of \$158,800. Even a 3-5% improvement in efficiency can generate \$44,800-\$76,800 in annual savings⁵.

3. Ergonomic Injury Rate and Workers' Compensation Incidents

Target improvement: Reduction in reported musculoskeletal incidents related to pushing, lifting, and repetitive bending—tracked against property baseline.

FlowCart's electric-assist base and modular linen handling reduce the most physically demanding motions in the cleaning workflow: pushing a loaded cart down corridors and lifting linen bundles. Reducing injury frequency lowers workers' compensation claims, sick days, and supervisory time spent managing injury cases. While this KPI requires baseline tracking to quantify precisely for each property, it is among the highest-impact quality-of-life improvements FlowCart delivers.

4. Supervisor Inspection Time per Floor

Target improvement: Reduce reactive floor-walking inspection time by 20-30% to redirect supervisor capacity toward higher-value tasks.

Currently, supervisors need to walk to each room after each housekeeper to inspect the quality of the work. This supervisory loop is manual, time-intensive, and dependent on his physical presence on the floor. FlowCart's task-tracking layer provides progress signals at the room level, allowing supervisors to prioritize inspections based on completion status rather than having to walk every floor.

5. Supply Waste and Restocking Frequency

Target improvement: Reduce mid-shift supply interruptions and generate supply usage data that improves purchasing accuracy over an established baseline.

Unnecessary trips from the floor to supply rooms are a direct source of time and travel waste. FlowCart's supply-aware module design keeps frequently needed items at the point of use, while usage monitoring can surface restocking needs before they cause workflow interruption. Over time, supply usage data also supports more accurate purchasing and inventory planning.

⁵ Detailed calculation can be found in Appendix.

Payback Period

Because the FlowCart subscription costs approximately \$4,200 per year per robot, the system can typically achieve payback within the first year of deployment, even under conservative efficiency improvements.

Future Revenue Streams

In addition to the core Robot-as-a-Service subscription, FlowCart creates opportunities for additional revenue streams as the platform expands. Future revenue may include specialized robotic modules designed for different hotel operational needs, such as high-turnover properties, large convention hotels, or premium housekeeping services.

The FlowCart system may also support advanced operational analytics tools, providing hotel managers with deeper insights into housekeeping workflows, supply usage, and labor allocation across properties. These analytics services could be offered through higher-tier software subscriptions.

05 Stage of Development of Product

Phase 1: Problem Validation and Workflow Research

FlowCart began with field research into hotel housekeeping operations, focusing on the everyday workflows surrounding the housekeeping cart. Through observation and workflow analysis (see Appendix, Figure 1), the team identified a key operational challenge: housekeepers frequently make repeated trips between guest rooms and service areas to transport supplies, dirty linens, and waste.

These repeated logistics movements slow down room turnover, increase physical strain on staff, and create clutter in guest corridors. The traditional housekeeping cart, while essential, concentrates all materials into a single large object that workers must maneuver across the entire floor.

This research led to the core concept behind FlowCart: distributing housekeeping logistics among smaller robotic modules rather than relying on a single large cart.

Phase 2: Prototype System Design

Based on these insights, the team designed FlowCart as a modular robotic housekeeping system (see Appendix, Figure 2) consisting of: a central smart cart platform, detachable modules for linens, supplies, and waste, and a software system that coordinates tasks and material movement. Our next prototype development focuses on: module mobility and navigation, modular attachment mechanisms, basic task coordination software, and human-centered interaction with housekeeping staff. This stage validates the technical feasibility of distributed housekeeping logistics. Prototype video can be accessed at <https://photos.app.goo.gl/euJSvmtUg5UAjx5o6>.

Phase 3: Pilot Deployment with Partner Hotels (our current stage)

The next phase focuses on real-world pilot deployments in selected hotel properties. Pilot programs allow the team to evaluate FlowCart within real housekeeping workflows and measure its impact on operational efficiency. Pilot testing will focus on: reduction in housekeeping travel distance, improvements in room turnover time, reduction in physical strain on staff, and operational feedback from hotel managers and housekeeping teams. These insights from pilot deployments will guide further refinement of both hardware and software.

Phase 4: Commercial Product Launch and Scaling

Following successful pilot validation, FlowCart will transition into commercial deployment across hotel groups and hospitality operators. Scaling efforts will focus on: expanding production capacity, integrating with hotel property management systems, developing additional specialized modules, and expanding deployments across multiple hotel properties. At scale, FlowCart becomes part of the operational infrastructure of modern hotels, supporting housekeeping logistics and providing real-time visibility into floor operations.

06 Competitor and Risk Analysis

Competitor Analysis

The competitive landscape for Nomu FlowCart spans two primary categories: traditional housekeeping carts and emerging robotic service platforms. While both serve important operational roles in hotels today, neither fully addresses the gap between physical workflow infrastructure and intelligent operational management.

Traditional Housekeeping Carts

The dominant incumbents in this category are established commercial manufacturers such as Rubbermaid Commercial Products and Carlisle FoodService Products, whose carts are widely deployed across hotels globally.

Traditional carts are low-cost (typically \$500-\$2,000), durable, simple to maintain, and fully embedded in hotel workflows. They require no technical training and pose no regulatory or IT risk. For decades, they have been treated as basic operational tools. However, they are passive infrastructure. They provide no intelligence, no data visibility, no environmental monitoring, and no integration with property management systems.

FlowCart builds directly on this familiar operational foundation, not by replacing the cart's function, but by transforming it from a passive tool into an intelligent operational node.

Robotic Service and Delivery Platforms

The second category consists of hospitality robots primarily designed for guest-facing delivery tasks. Companies such as **Relay Robotics**, **Savioke**, and **Pudu Robotics** have introduced autonomous robots capable of delivering amenities, food, or small items to guest rooms. These systems typically cost \$15,000-\$25,000+ per unit and emphasize autonomy and elevator integration. They represent important technological advancements and demonstrate that hotels are open to robotics adoption.

However, these platforms are not designed to support housekeeping workflow. They operate as task-specific delivery devices rather than embedded operational infrastructure. Their integration depth with housekeeping management systems is limited, and they do not function as mobile sensing or workflow coordination tools. In many properties, they remain novelty or marketing enhancements rather than core operational systems.

Market Gap and Strategic Positioning

FlowCart occupies the strategic gap between these two categories. Traditional carts are deeply integrated into daily operations but lack intelligence. Robotic delivery platforms offer automation but are disconnected from housekeeping workflows, which constitute the largest labor segment in hotels.

FlowCart integrates directly into the existing housekeeping ecosystem while adding real-time data collection, environmental sensing, workflow visibility, and modular automation capabilities. Instead of introducing a standalone robot, FlowCart transforms the most ubiquitous and overlooked piece of hotel infrastructure into a distributed operational intelligence platform.

This positioning offers several strategic advantages:

- Lower adoption barrier compared to full robot replacement
- Native alignment with housekeeping workflows
- Clear efficiency tracking and supply optimization
- Potential integration with property management and facilities systems

Rather than competing on novelty or full autonomy, FlowCart competes on operational leverage. Hotels do not necessarily need more lobby robots. They need better visibility and intelligence in the areas where the real operational work occurs. FlowCart turns the housekeeping cart from a cost center into a management asset.

Risk Analysis

While FlowCart addresses a clear operational opportunity, risks must be considered:

1. Technology Integration Risk: Hotels operate complex technology ecosystems, including PMS systems, housekeeping management software, and building

infrastructure. Integration with these systems may require additional development and coordination with third-party vendors.

2. Operational Adoption Risk: Housekeeping teams rely on familiar workflows and tools. If new technology disrupts daily routines or increases complexity, adoption may be slower than expected. FlowCart must therefore prioritize intuitive design and seamless integration into existing processes.

3. Cost Sensitivity in Hospitality: The hospitality industry is highly cost-sensitive, particularly in midscale segments. FlowCart must demonstrate clear return on investment through measurable improvements in labor efficiency, reduced injuries, or faster room turnover;

4. Competitive Expansion: As robotics adoption increases in hospitality, existing robotics companies may expand into operational workflow products. Large robotics firms or established hotel suppliers could attempt to introduce competing solutions.

Risk Mitigation Strategy

Nomu mitigates these risks through several strategic design choices:

- Workflow-native design, ensuring compatibility with existing practices;
- Modular deployment, allowing hotels to pilot the system before scaling;
- Integration readiness with major PMS and operational software platforms;
- Clear ROI metrics tied directly to labor efficiency and operational visibility.

Therefore, we position Nomu FlowCart as an augmentation of existing infrastructure rather than a full replacement of labor or workflow systems.

07 Management Team



Nanyi Jiang - CEO | Business, Design, Human-Robot-Interaction

Nanyi Jiang leads FlowCart with a background spanning business strategy, design research, and HRI. She is a Cornell InfoSci PhD student focusing on research in human-robot interaction and design. Previously, she had worked at IDEO, Ford Innovation Lab, and two startups. She earned her MFA in Media Design from ArtCenter College of Design and her BA in Economics from UCLA.



Borui Wang - CPO | Product, Design, Prototyping

Borui leads product strategy and execution at FlowCart, overseeing the design, prototyping, and iteration of the FlowCart platform. A PhD in Information Science and a strong background in product design and rapid prototyping, he translates operational and user insights into scalable, manufacturable product solutions.



Michael Sack - CTO | Design, HRI, Engineering

Michael leads user research and engineering for the FlowCart platform. As a PhD Candidate with an MS in Information Science, he has a background in cognitive science, human-robot interaction, and design research. His work centers on the end-to-end research, design, and integration process of collaborative autonomous systems across various work contexts.



Isabella An - CSO | Strategy, Finance, Hospitality Operations

Isabella leads strategy at FlowCart, grounding product development in the operational and financial realities of hospitality. She is a student at Cornell University's School of Hotel Administration with a focus on finance and operations. Her work centers on market validation, unit economics, and go-to-market strategy.



Mahin Hardaway - COO | Hospitality Operations, Business Development

Mahin leads operations at FlowCart, driving product execution and strategic expansion across the hospitality industry. He is an undergraduate student at Cornell University's School of Hotel Administration, specializing in analytics, operations, and entrepreneurship. His work focuses on operational integration, product refinement, and market expansion.

Advisors

Malte Jung — Cornell Associate Professor | HRI in workplace expert

Malte Jung is an Associate Professor in Information Science at Cornell University, focusing on design and behavioral aspects of human-robot interaction in group and team settings. His research explores how autonomous systems become integrated into everyday activities and work, making him a valuable strategic advisor on FlowCart's human-centered robotics behavior and workplace collaboration.

Leila Takayama — VP Design in Robust AI | HRI & Research to product expert

Leila A. Takayama is the VP of Design at Robust AI, an intelligent warehouse robot startup. She was an Associate Professor of Human-Computer Interaction at the University of California, Santa Cruz, and a former manager of human-robot interaction at Google X and Willow Garage.

Ron Swidler — CEO of The Gettys Group | Hospitality Innovation

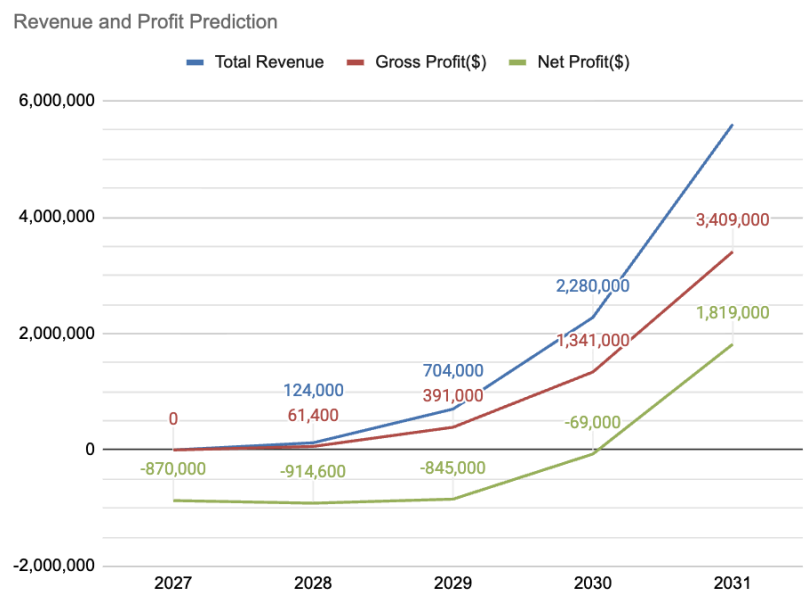
Ron Swidler is the CEO of The Gettys Group, a global hospitality design, branding, and consulting firm focused on shaping the future of hotels and guest experiences. With decades of leadership in hospitality strategy and development, he is known for driving innovation and applying future technologies to hospitality operations and service design.

Michael Llodra — Managing Director, AXCS | Business Strategy & Finance

Michael Llodra is the Managing Director at AXCS Investment, where he focuses on investment strategy, business growth, and financial planning. He brings extensive experience in evaluating, structuring, and scaling technology-driven ventures.

08 Financial Statements

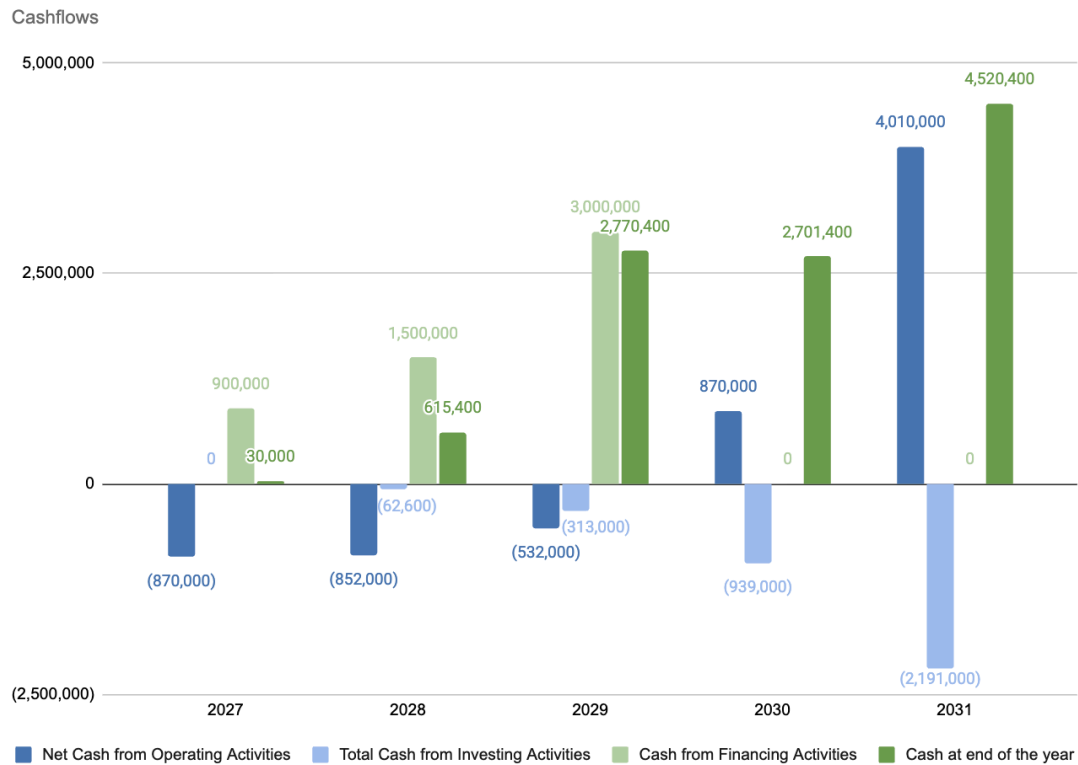
Income⁶



We view 2026 as year 0, when we will continue research and develop our prototype using the initial funding. Year 1 (2027) will be our beta prototyping; therefore, no actual robot will be deployed. In 2028, we predict the deployment of 20 new robots. In year 5, 2031, we will have 1,000 deployed robots, generating a net profit of \$1.82 million.

⁶ The graph is derived from our income statement, there will be details in Appendix

Cash Flow⁷



Balance Sheet

Metric	Year 1	Year 2	Year 3	Year 4	Year 5
ASSETS					
Current Assets					
Cash and cash equivalents	30,000	615,400	2,770,400	2,701,400	4,520,400
Total Assets	30,000	615,400	2,770,400	2,701,400	4,520,400
LIABILITIES					
Accounts payable	0	0	0	0	0
Loans	0	0	0	0	0
Total Liabilities	0	0	0	0	0
EQUITY					
Paid-in capital (Pre-seed, Seed, Series A)	900,000	2,400,000	5,400,000	5,400,000	5,400,000
Retained earnings (accumulated losses/profits)	(870,000)	(1,784,600)	(2,629,600)	(2,698,600)	(879,600)
Total Equity	30,000	615,400	2,770,400	2,701,400	4,520,400
Total Liabilities + Equity	30,000	615,400	2,770,400	2,701,400	4,520,400

⁷ The graph is derived from our Cashflow Statement, details in Appendix

09 Offering of the Company

Nomu is seeking initial capital to develop and deploy the FlowCart platform through Initial Funding, pre-seed, seed, and Series A rounds.

Initial Funding: Alpha Prototype Development (\$25,000)

The competition prize will be used to support the development of a fully-functional alpha prototype to test our ideas. We will be able to work on: refinement of FlowCart modules; iterative engineering, development, and testing; preparation for pilot deployments with hotel partners; and initial legal and business formation expenses. This funding enables the team to move from our current, not fully functional prototype to a fully functional prototype and a pilot-ready system.

Pre-Seed Round (Target raise: \$900,000)

The pre-seed round will focus on building the first operational prototype and validating FlowCart in real hotel environments. The goal of the pre-seed stage is to demonstrate technical feasibility and measurable operational impact through pilot deployments. Primary uses of capital include: hardware prototyping and engineering development, software platform development, pilot deployments in more partner hotels, user research with housekeeping teams, and initial hiring for engineering and product development.

Seed Round (Target raise: \$1.5 million)

The seed round will support the transition from pilot deployments to early commercial rollout. Funding will be used to: refine the production version of FlowCart; expand pilot deployments into multi-property trials; develop manufacturing partnerships and supply chains; expand the software platform and operational analytics; and build initial sales and business development capacity. By the end of the seed stage, Nomu aims to establish FlowCart as a validated operational technology within the hospitality industry.

Series A (Target raise: \$3 million)

Series A funding will support scaling production and expanding deployments across hotel portfolios. Key investment areas include: scaling hardware manufacturing, expanding sales and industry partnerships, integrating FlowCart with hospitality management systems, developing additional robotic modules and software features, and expanding deployments across hotel chains and management groups. At this stage, FlowCart transitions from pilot technology to scalable operational infrastructure for the hospitality industry.

Appendix

Appendix

Calculation for TAM/SAM/SOM

TAM	\$19B/year	U.S. housekeeping labor spend 1.3B occupied room nights (U.S.) CPOR (housekeeping labor) = \$44.62 → $1.3B \times 44.62 = \$58B/\text{year}$ (U.S.) Global scale using room ratio U.S. rooms $\approx 5.3M$, Global rooms $\approx 17.5M$ Ratio $\approx 17.5 / 5.3 \approx 3.3\times$ Global housekeeping labor $\approx 58B \times 3.3 \approx \$190B/\text{year}$ Flowcart 10% automation assumption TAM: $190B \times 10\% \approx \$19B/\text{year}$
SAM	\$3.77B/year	We assumed: 65% of U.S. hotel rooms are in segments likely to adopt robotics (midscale + upper-midscale + upscale) U.S. automation-adjusted TAM = $58B \times 10\% = \$5.8B$ 65% of that = $5.8B \times 0.65 = \$3.77B$
SOM	\$40.6M/year	Focus on upper-midscale + upscale chains (~35% of U.S. rooms), Assume 2% early adoption over 5 years $5.8B \times 35\% \times 2\% = 5.8B \times 0.35 \times 0.02 = \$40.6M/\text{year}$

Product Development

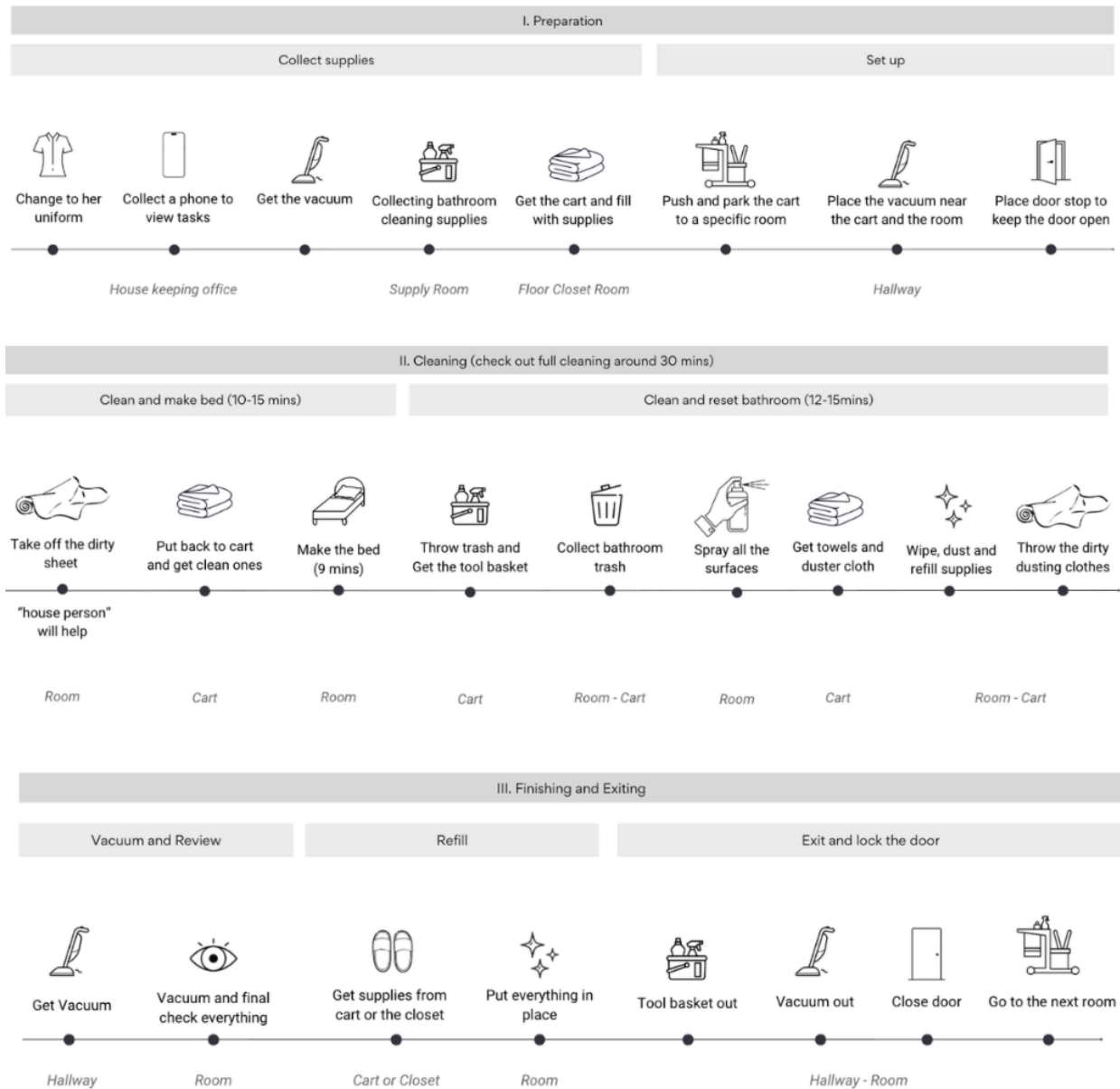


Figure 1: Analysis of Housekeeping Workflow from Observation

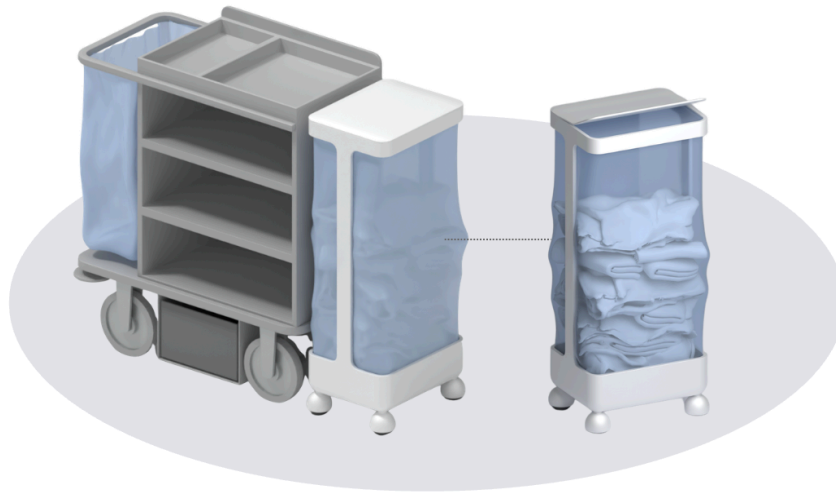


Figure 2: Example of designing of prototype

Labor Saving Calculation

Saving Rates (%)	Labor Annual Saving(\$)	Net benefit (\$ per year)
3	49,000	44,800
5	81,000	76,800
10	163,000	158,800

Current Housekeeping labor cost $\approx$ \$44.62 per occupied room(including room attendants+houseman+laundry staff+supervision+benefits).

The U.S. hotels average about 67% occupancy.

$\sim 150\text{-rooms} \times 67\% \text{ occupancy} \approx 100$ rooms cleaned per day

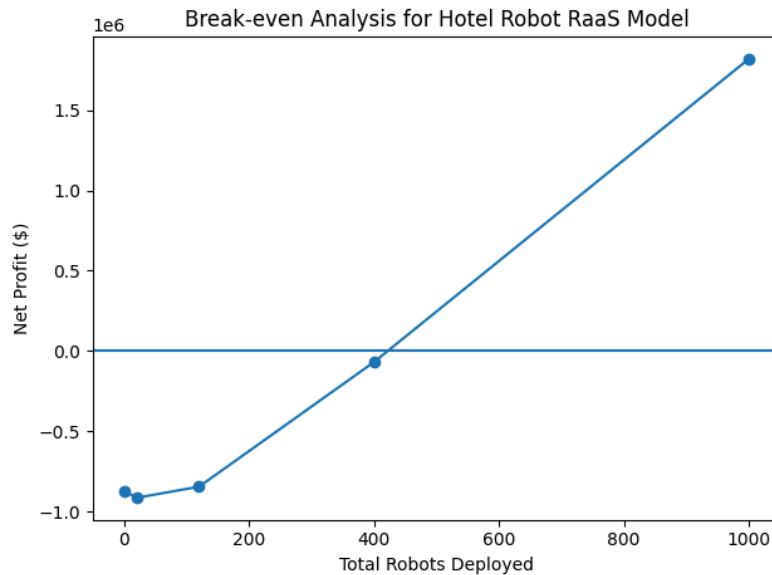
Daily Cleaning Cost: $100 \text{ rooms} \times \$44.62 \approx \$4,462$ per day

Annual Cleaning Cost: $\$4,462 \times 365 \approx \1.63M per year

Income Statement

Metric	▼	#	2027	▼	#	2028	▼	#	2029	▼	#	2030	▼	#	2031	▼
New Robot Deployed			0			20			100			300			700	
Total Robots Deployed			0			20			120			400			1,000	
Setup Fee per Robot			2,000			2,000			2,000			2,000			2,000	
Yearly Subscription per Robot			4,200			4,200			4,200			4,200			4,200	
Total Revenue (\$)			0			124,000			704,000			2,280,000			5,600,000	
Manufacturing + components			2,200			2,200			2,200			2,200			2,200	
Shipping + Tariff			410			410			410			410			410	
US Assembly			400			400			400			400			400	
Delivery			120			120			120			120			120	
COGS per Robot (\$)			3,130			3,130			3,130			3,130			3,130	
Total COGS (\$)			0			62,600			313,000			939,000			2,191,000	
Gross Profit (\$)			0			61,400			391,000			1,341,000			3,409,000	
Salaries			530,000			530,000			760,000			850,000			850,000	
R&D			200,000			200,000			200,000			200,000			200,000	
Sales and Marketing			100,000			100,000			100,000			100,000			100,000	
Operation and Logistics			0			100,000			100,000			100,000			100,000	
Cloud Infrastructure and Software			40,000			40,000			40,000			40,000			40,000	
Maintenance and Field Services			0			6,000			36,000			120,000			300,000	
Total Operating Expenses (\$)			870,000			976,000			1,236,000			1,410,000			1,590,000	
Net Profit (\$)			-870,000			-914,600			-845,000			-69,000			1,819,000	

Break-Even



Based on projected deployment and operating costs, the company reaches break-even when approximately 450 robots (approximately 60 - 90 150-room hotels) are deployed in the field. At this scale, recurring subscription revenue from the installed robot base is sufficient to cover operating expenses. Beyond this point, additional robot deployments generate increasing profit due to the high-margin subscription model.

Cash Flow Statement

Metric	▼	#	2027	▼	#	2028	▼	#	2029	▼	#	2030	▼	#	2031	▼
Operating Activities																
Cash receipts from customers			0			124,000			704,000			2,280,000			5,600,000	
Cash paid for salaries			(530,000)			(530,000)			(760,000)			(850,000)			(850,000)	
R&D expenses			(200,000)			(200,000)			(200,000)			(200,000)			(200,000)	
Sales & marketing			(100,000)			(100,000)			(100,000)			(100,000)			(100,000)	
Operations & logistics			0			(100,000)			(100,000)			(100,000)			(100,000)	
Cloud infrastructure			(40,000)			(40,000)			(40,000)			(40,000)			(40,000)	
Maintenance & field services			0			(6,000)			(36,000)			(120,000)			(300,000)	
Net cash from operating activities			(870,000)			(852,000)			(532,000)			870,000			4,010,000	
Investing Activities																
Robot manufacturing investment			0			(62,600)			(313,000)			(939,000)			(2,191,000)	
Total cash from investing activities			0			(62,600)			(313,000)			(939,000)			(2,191,000)	
Financing Activities																
Pre-seed funding			900,000									0			0	
Seed funding			0			1,500,000			0			0			0	
Series A funding			0			0			3,000,000			0				
Total cash from Financing Activities			900,000			1,500,000			3,000,000			0			0	
Cash at beginning of year			0			30,000			615,400			2,770,400			2,701,400	
Total Change in Cash			30,000			585,400			2,155,000			(69,000)			1,819,000	
Cash at end of the year			30,000			615,400			2,770,400			2,701,400			4,520,400	